



2026 SUMMER SCHOOL

Transforming Technology, Modern Practice, and Clinical Impact

JUNE 16–20 | UNIVERSITY OF MICHIGAN

PET-Guided Adaptive Radiotherapy

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Disclosure

- Research funding from RefleXion Medical, Inc.



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Foundations and Motivations

- The role of PET in Radiation Oncology
 - Evolution from Anatomical Imaging to Biological Functional Imaging
 - Reveals tumor metabolism, hypoxia, and cellular activity for better treatment targeting
 - Biology-guided Radiotherapy (BgRT)
 - Enables dynamic radiation dose adjustments based on evolving tumor biology during treatment
 - Definition of PET-guided Adaptive Radiotherapy
 - PET-derived information is incorporated to modify treatment planning or delivery
 - Offline (replanning between fractions based on interim PET scans)
 - Online (dynamic fluence adjustment during beam delivery)
 - Supports personalized treatment potentially improving tumor control and reducing toxicity

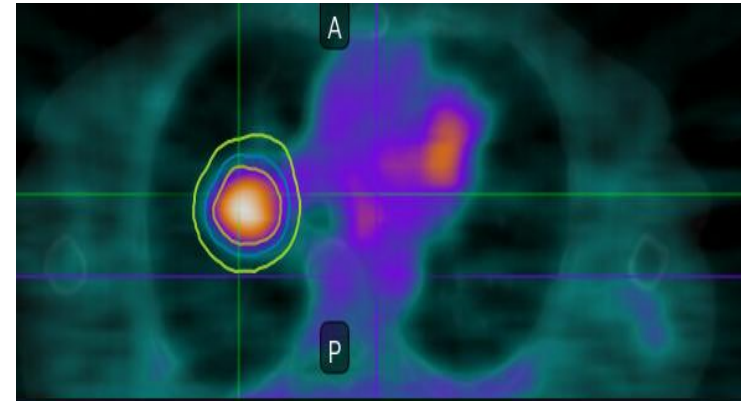
Clinical Quantitative PET metrics

Diagnostics

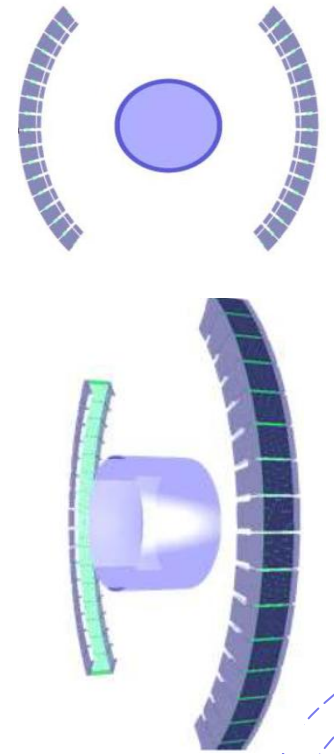
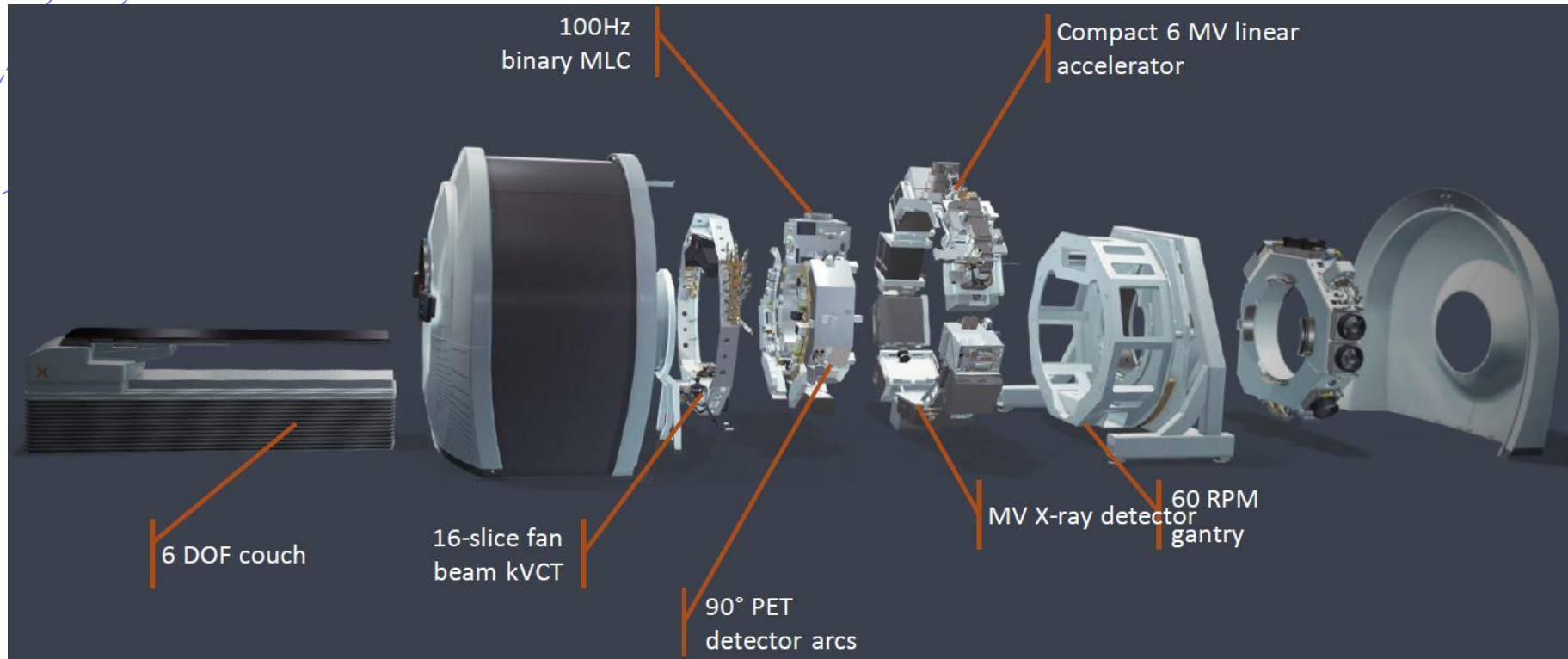
- SUVmax (Max Standardized Uptake Value): measures the highest metabolic activity in a tumor
- SUVmean (mean Standardized Uptake Value): Average metabolic activity within a defined region
- MTV (Metabolic Tumor Volume): quantifies the volume of tumor tissue exhibiting increased FDG uptake
- TLG (Total Lesion Glycolysis): A composite metric combining MTV and SUVmean, reflecting both tumor volume and metabolic activity

Radiotherapy

- **AC** (Activity Concentration)
- **NTS** (Normalized Target Signal)

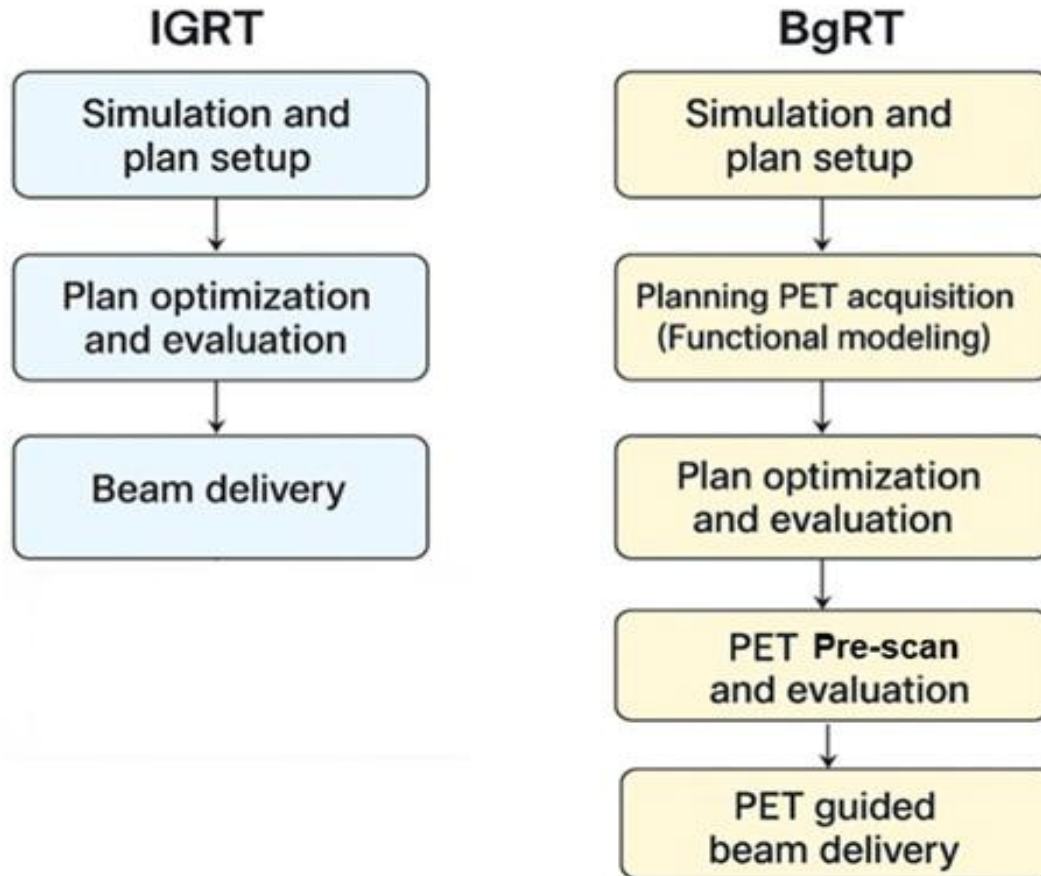


Methods and Clinical Integration



RefleXion X1 PET-Linac components (Courtesy of RefleXion Medical)

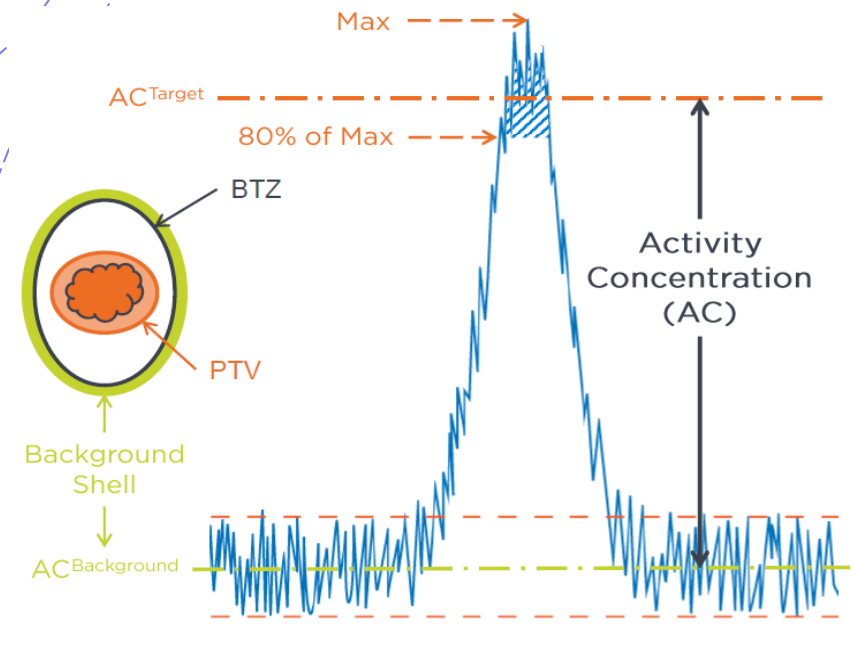
RefleXion SCINITIX System and Workflow



Real-Time PET-Guided Delivery

Tumor PET signals act as **biological fiducials** guiding beam modulation during treatment for high precision therapy.

Activity Concentration (AC) & Normalized target signal (NTS)



- Target AC = AC_{Target} - AC_{Background}
- Evaluated at treatment planning & Pre-Tx delivery
- Mandatory thresholds at planning and delivery for RefleXion X1 :

$$AC \geq 5\text{kBq/ml}$$

$$NTS = \frac{AC_{\text{Target}} - AC_{\text{Background}}}{\sigma_{\text{Background}}}$$

$$NTS \geq 2.7 \text{ at planning}$$
$$NTS \geq 2.0 \text{ at delivery}$$



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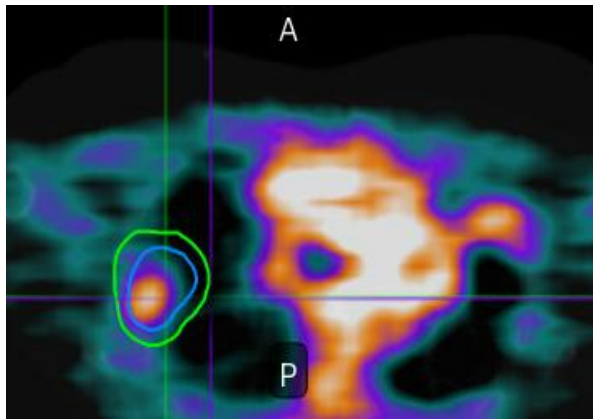
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Non-clinical BgRT PET parameters (TC & TCR)

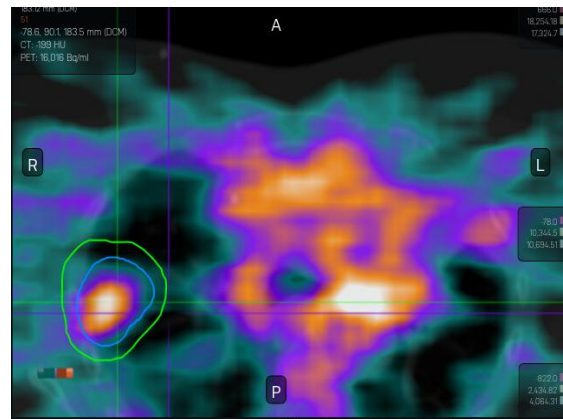
- Target Contrast at Functional Modeling (TC_FM):

$$\frac{\text{mean PET activity within the PTV} - \text{mean PET activity within BTZ excluding PTV}}{\text{mean PET activity within the PTV}}$$

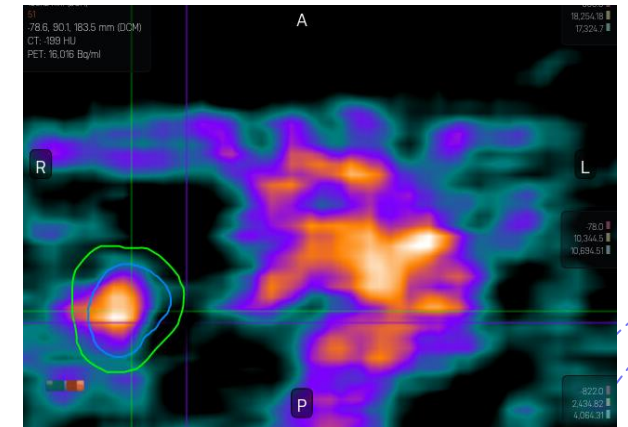
- Target Contrast Ratio during treatment (TCR): $\frac{\text{TC at pre-TX}}{\text{TC at FM}}$



FM/planning



Pre-treatment

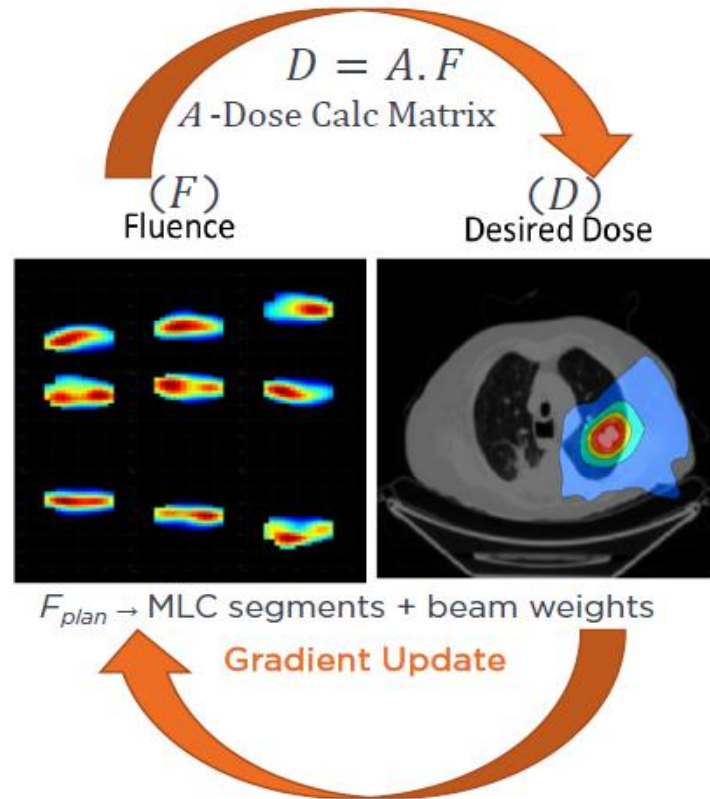


During treatment

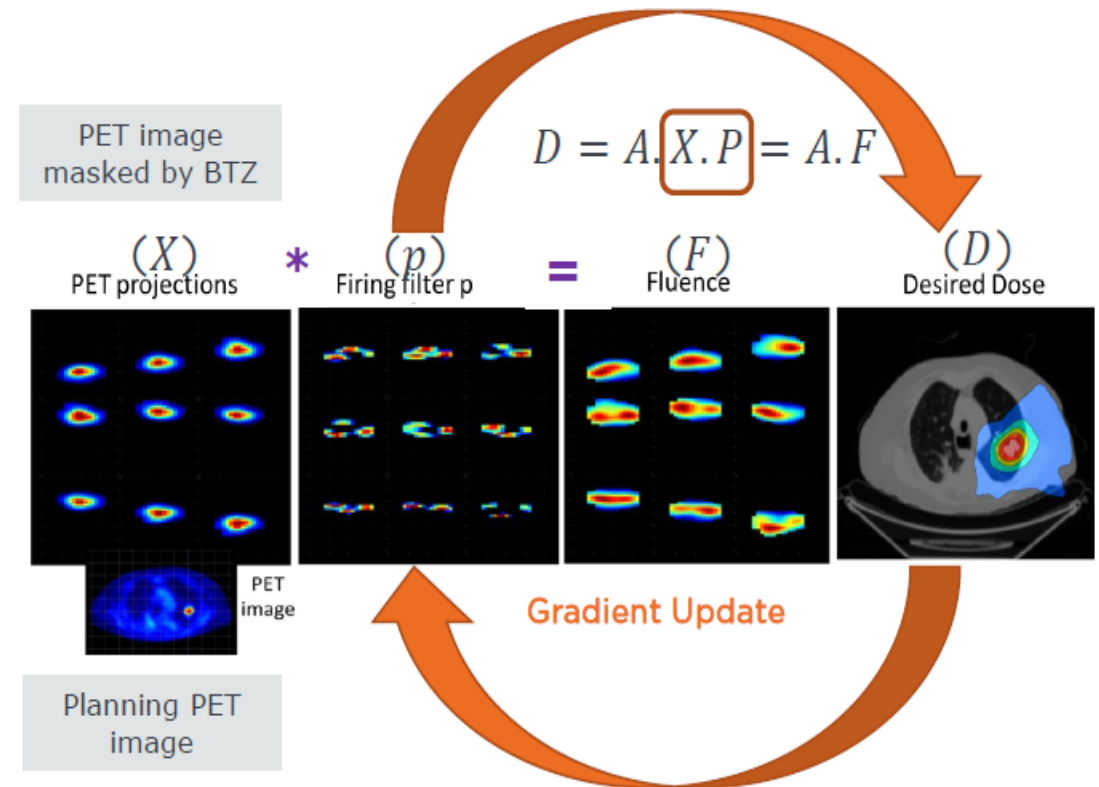
(Blue contour: PTV, Green contour: BTZ)

SBRT vs. BgRT: treatment planning

SBRT Dose Calculation:
Solve for fluences that meet dose constraints



BgRT Dose Calculation:
Solve for the optimal firing filters that meet dose constraints

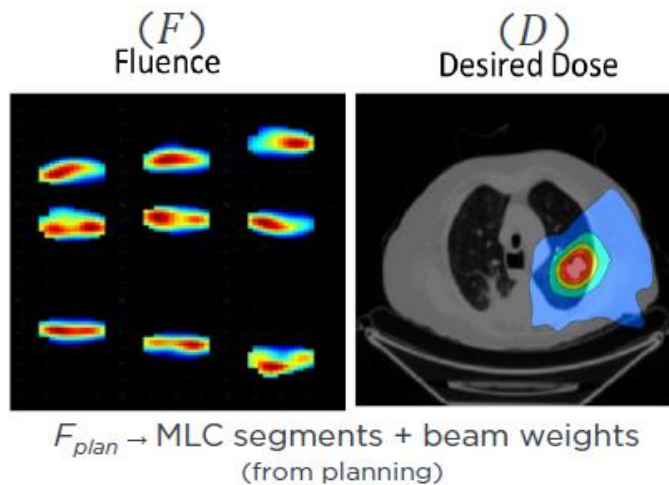


(Courtesy of Reflexion Medical)

SBRT vs. BgRT: treatment delivery

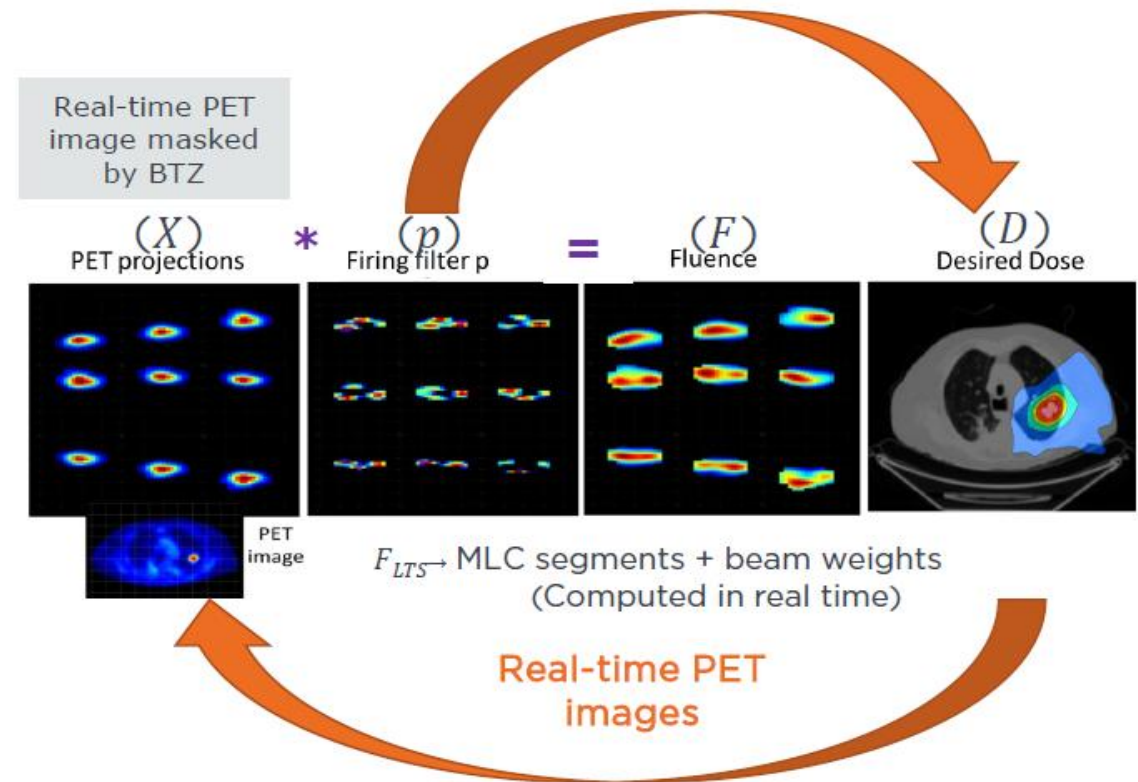
SBRT Delivery

Fluence is pre-computed at planning



BgRT **Tracked-dose** Delivery

Fluence is updated in real time from LTS images



(Courtesy of Reflexion Medical)

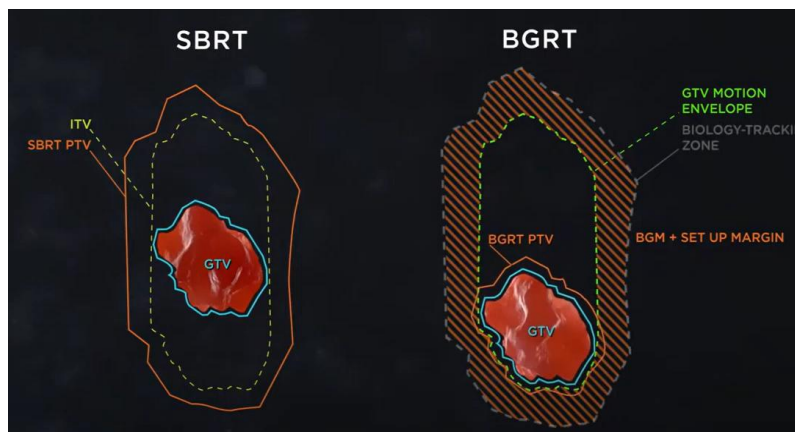
SCINTIX Therapy (BgRT) numbers at Yale

as of early June, 2026

(patient numbers)	FM/Simulation	Tx (started)	Tx (completed)
Lung	20	13	9
Bone	3	2	1
Total	23	15	10

Total sessions numbers (functional modeling + SCINTIX Tx) = 25+53=78

Initial Yale experience: first 15 BgRT candidates with tumor characteristics (GTV/PTV/motion range)

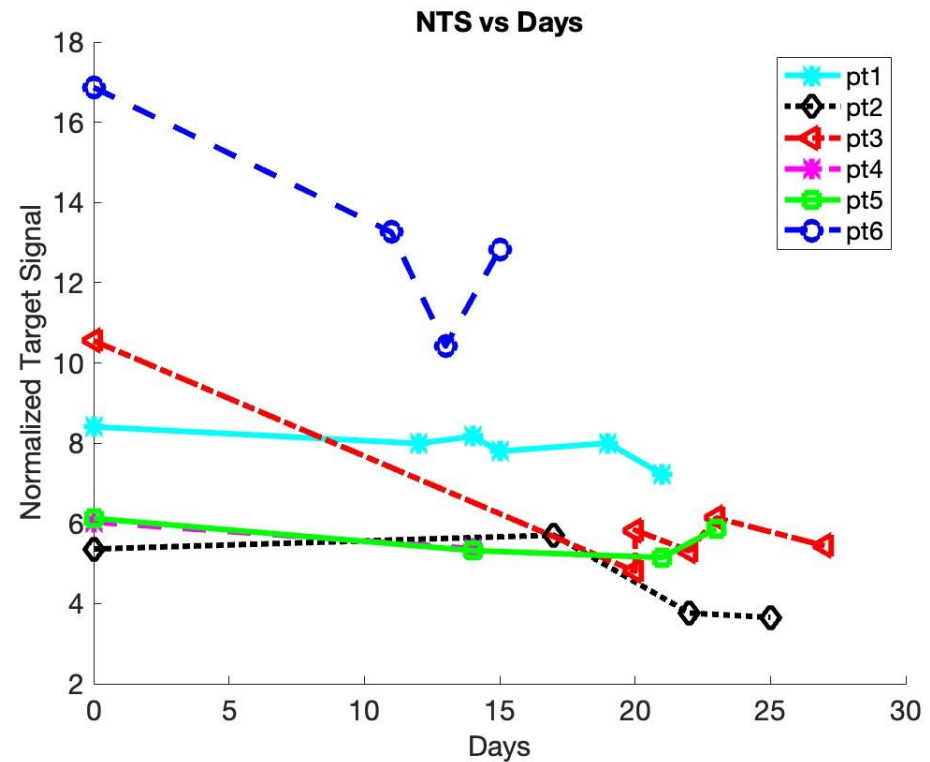
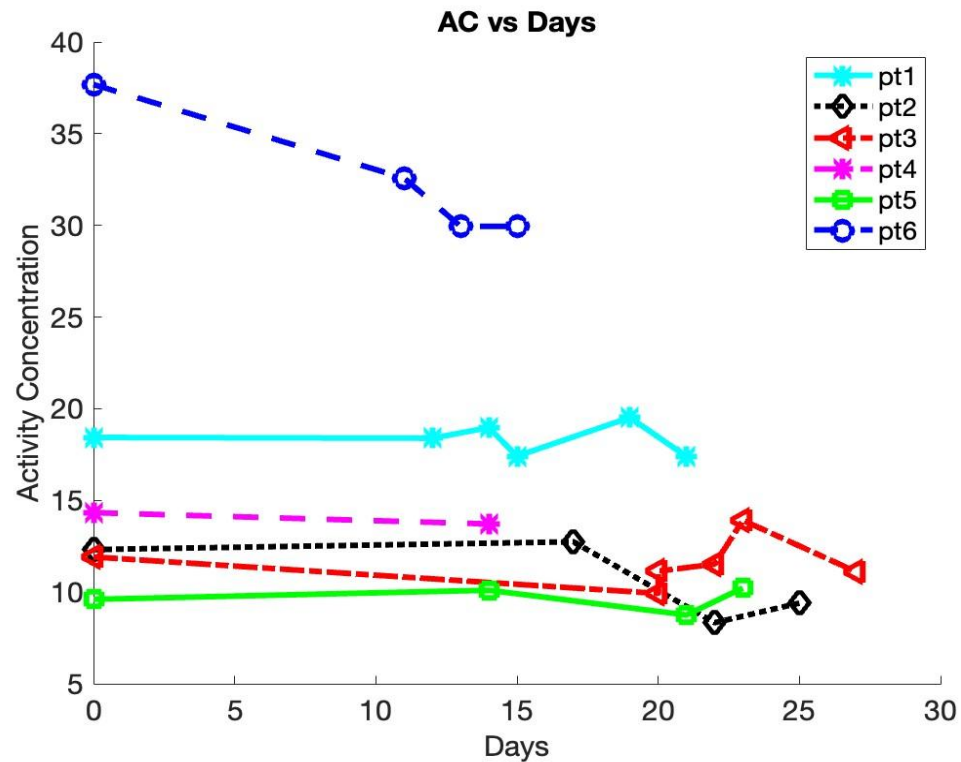


SBRT Vs. BgRT: GTV, ITV, PTV, BTZ

Case	GTV (cc)	Motion (cm)	BgRT PTV (cc)	SBRT PTV (cc)	% Volume Reduction
1	1.2	0.6	8.8	13.3	34
2	1.8	1.5	18	30.2	40
3	8.4	1.8	38.5	66.8	42
5	21.1	0.7	67.7	86.4	22
6	7.1	0.2	22.2	26.4	16
7	5.6	0.2	18.4	22.1	17
9	5.8	0.1	12.3	19.8	38
10	13.3	0.5	36.1	40.5	11
11	3.8	0.5	15	21	29
12	4.8	0.1	16.5	18.2	9
13	19.4	0.6	50.4	64.2	21
14	1.6	0.5	7.7	14.09	45
15	3.7	1.5	14	29.5	53

Lung target motion & reduction in PTV volume using BgRT versus SBRT

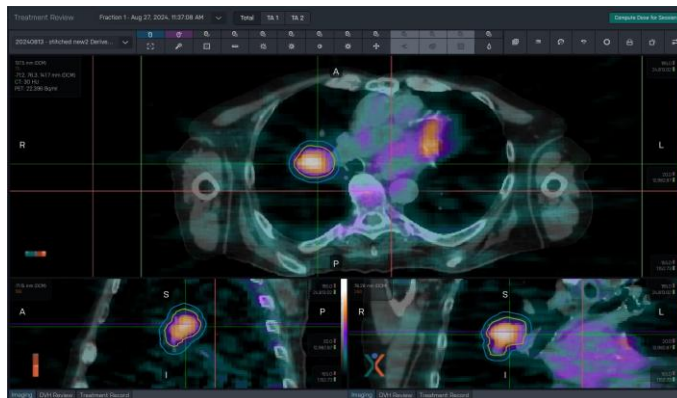
Initial Yale experience: summary of PET metric (AC and NTS) of 6 BgRT patients (total of 25 fractions)



- The first data point at day 0 is from the PET data acquired during Functional Modeling
- The subsequent data points represent each BgRT treatment fraction

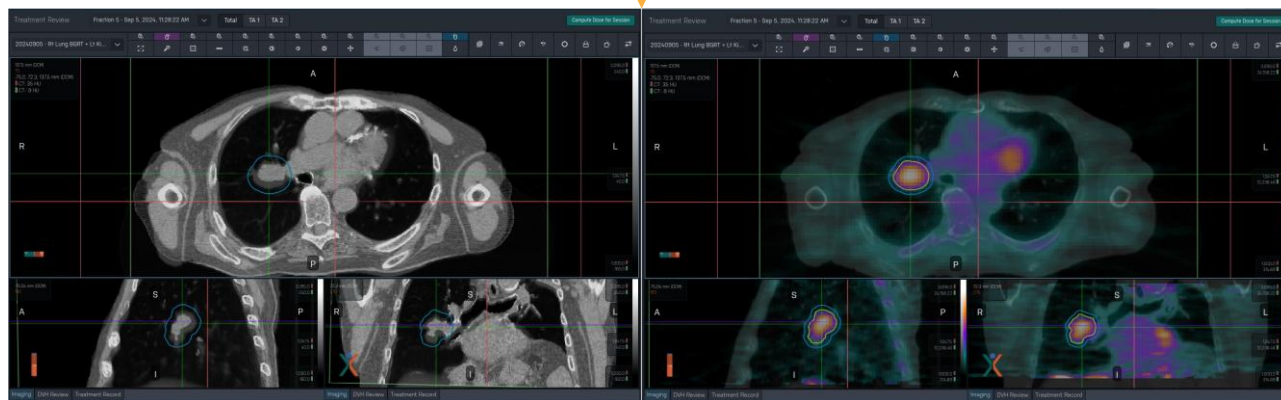
First Single-plan, Multi-Modality Radiotherapy Tx (MTT hybrid: BgRT/Rt Lung + IGRT/Lt Kidney)

1st Tx fraction: →



- **Oligoprogressive 3.8 cm Rt ultra-central lung met of thymic origin ($SUV_m=9.3$)**
- **Oligoprogressive 2.4 cm Lt kidney**
 - 40 Gy in 5 fractions to both lesions
 - Multiple modalities in a single plan
 - Two treatment areas jointly optimized
 - One patient setup with separate CT localization

5th Tx fraction: tumor shrinkage observed PET signal still aligned:

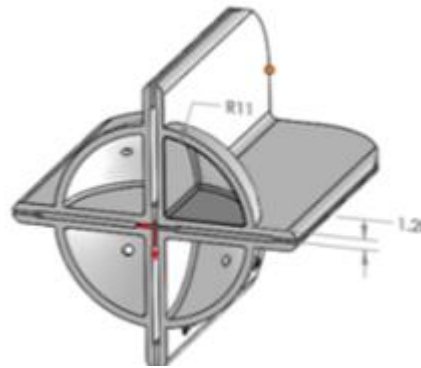


- **MTT delivered successfully with high conformality**
 - Synthetic PET imaging: lung tumor met BgRT criteria ($AC \geq 5$ kBq/ml, $NTS \geq 2.7$), confirmed on functional modeling PET
 - Target coverage stayed at 100% meeting all dose constraints
 - Average treatment time per fx ~36 min for BgRT and ~16 min for SBRT (52 min total)

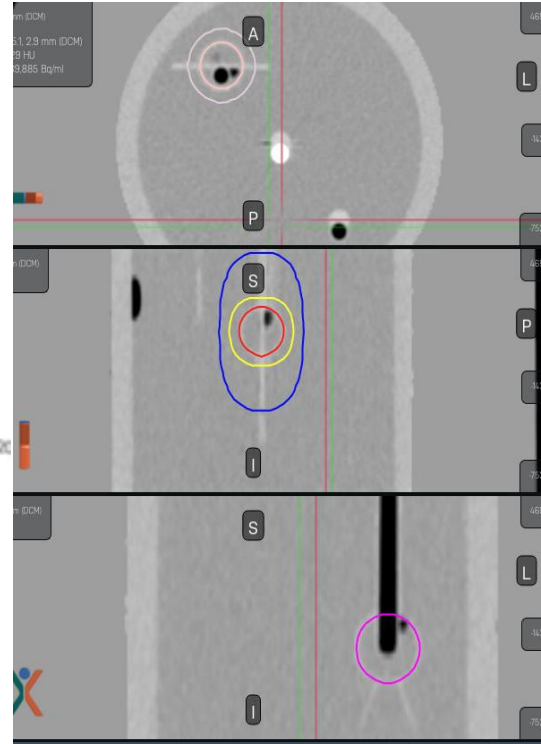
Phantom Studies for Multi-Target PET-guided Radiotherapy



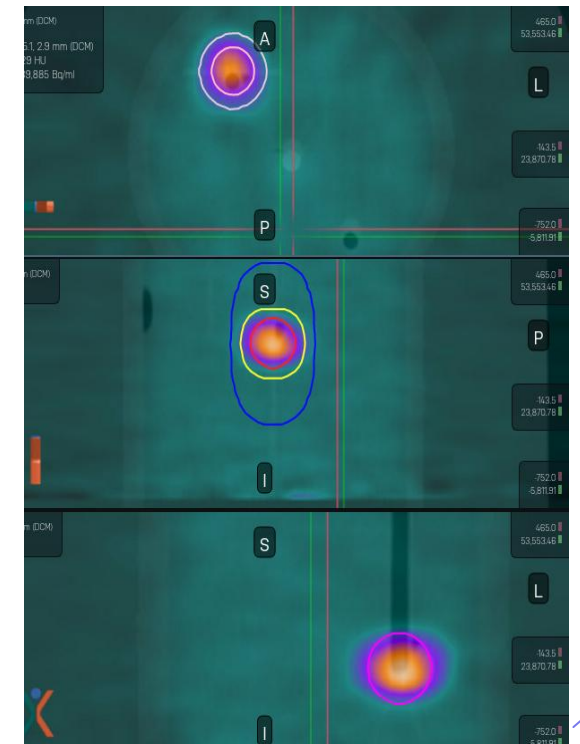
ArcCheck inserts
(c shape or spherical ball)



22mm spherical target

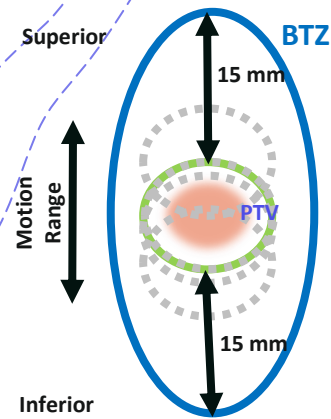


planning CT

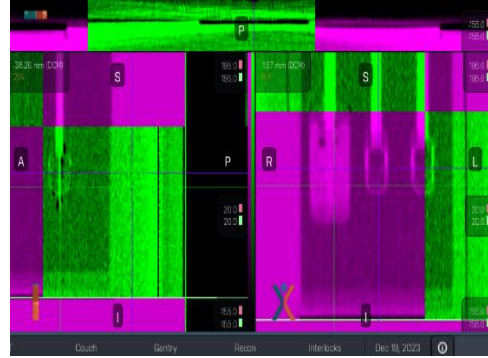


planning PET

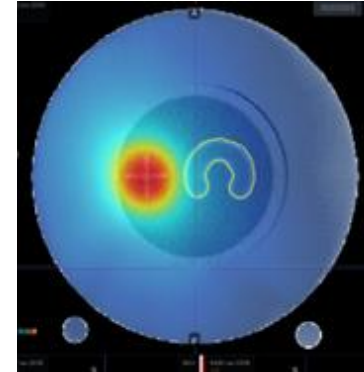
Sample Motion Experiment



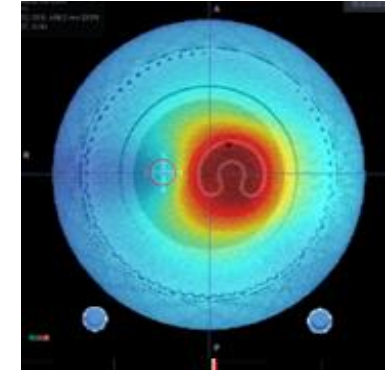
Sample target contouring



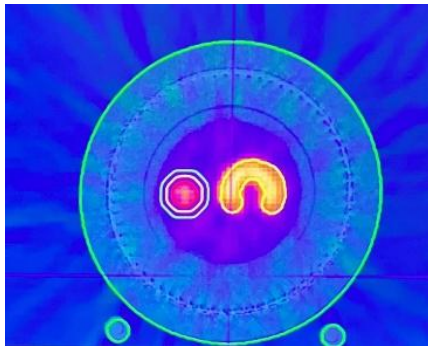
KVCT localization



Plan 1



Plan 2

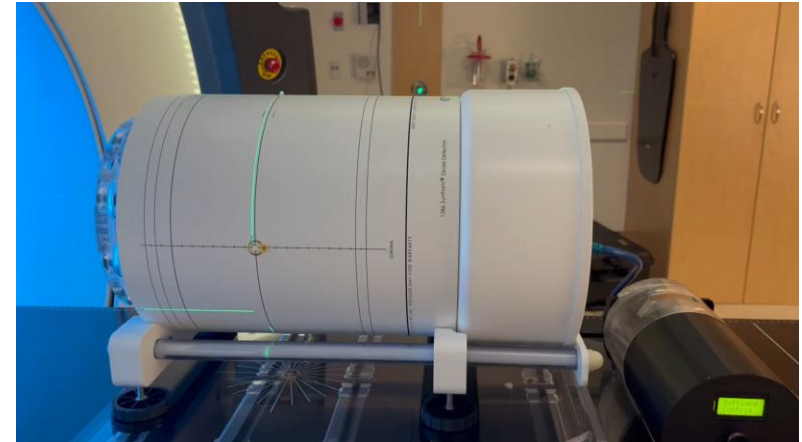
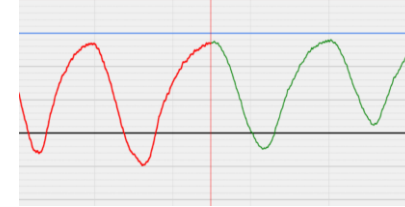


Sample PET image

1. Average motion CT was used as Reference and Planning CT
2. Mid-respiration GTV was used for planning
3. GTV was at the center of the BTZ, with 15 mm margin in the superior and inferior direction. This ensured full coverage of the motion range (15mm)
4. For functional modeling (and treatment), the phantom was setup to the mid-respiratory phase, and motion was on start from kVCT acquisition

BGRT session in minutes for single target phantom testing

- FDG injection to PET imaging/uptake **(75~80mins)**
 - Phantom set up (~15mins)
 - KVCT localization (5~10mins)
 - Approval of KVCT localization (~3 mins)
if needed, second KVCT after phantom alignment adjustment
 - Planning PET (~20mins) or Pre-delivery PET acquisition (~6mins)
 - PET evaluation for delivery (~6mins)
 - Sign-off for BGRT delivery (~1min)
 - BGRT delivery **(25-30 mins)**
 - Post treatment dose calculation (~2 mins)
- Total phantom on table time **(59~70mins)**



Motion platform with patient breath wave input

Clinical Evidence and Ongoing Trials

Trial name	Clinical trial number	Brief description
PREMIER: Prospective Data Registry and Quality of Life Assessment of Patients Undergoing Radiotherapy With the RefleXion Medical Radiotherapy System	NCT05406167	A prospective registry assessing outcomes, quality of life, and economic impact in patients treated with RefleXion X1 (BgRT, SBRT, IMRT)
BIOGUIDE-X2: Performance and Safety of Biology-guided Radiotherapy Using the RefleXion Medical Radiotherapy System in a Variety of Solid and Soft Tissue Tumors	NCT06634706	<i>In silico study</i> evaluating safety and performance of BgRT using FDG-PET in new anatomic sites such as Head & Neck, Pelvis, Retroperitoneal, and Hepatobiliary
A Pilot Prospective Study of the RefleXion [18F]- FDG PET-CT Subsystem Imaging Performance in Patients With Various Malignancies	NCT05844306	Investigation of the image quality and technical accuracy of the RefleXion X1 FDG PET-CT subsystem in the Abdomen, Pelvis, Thorax, H&N, and Breast compared to standard PET-CT
Nonsignificant Risk (NSR) Study of [68Ga]-PSMA-11 (Illuccix) as a BgRT BioGuide on the RefleXion X1 System in Patients Already Undergoing Diagnostic PET (Positron Emission Tomography) PSMA (Prostate Specific Membrane Antigen) Imaging	NCT06353321	A feasibility study exploring the use of [68Ga]-PSMA-11 as a Prostate radiotracer for BgRT in prostate cancer
A Prospective Study of the RefleXion [18F]- DCFPyL PET-CT Subsystem Imaging Performance in Patients With Prostate Cancer	NCT05470699	A feasibility study exploring the use of [18F]-DCFPyL as a radiotracer for BgRT in prostate cancer



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Summary of BgRT tracers

Tracer (radiopharmaceutical)	Indication	Role In BgRT	Literature
¹⁸ F-FDG	Lung & bone tumors (primary/metastatic)	Standard tracer for real-time PET-guided BgRT (FDA-cleared for SCINTIX BgRT in lung & bone cancers)	Park et al. 2025 Chen et al. 2025
⁶⁸ Ga-PSMA/ ¹⁸ F-PSMA	Prostate cancer with bone metastases	Feasibility demonstrated for SCINTIX BgRT treatment in metastatic prostate cancer	Hrinivich et al. 2019 Han et al. 2024
¹⁸ F-FMISO	Lung and Head Neck	Supports dose painting to radioresistant subregions and monitoring of reoxygenation	Kelada et al. 2018
¹⁸ F-FES (Fluoroestradiol)	ER+ breast cancer lesions	Pilot study comparing FES and FDG PET metrics for BgRT eligibility in breast cancer	Gibbard et al. 2025
⁶⁴ Cu-ATSM	Recurrent malignant glioma	Studied for hypoxia-guided dose painting in BgRT-like planning (study protocol for a phase III trial)	Ando et al. 2025
FAPI (Fibroblast activation protein inhibitor-based)	Various epithelial cancers	Promising for tumor microenvironment- guided dose targeting; ongoing dosimetry and biodistribution studies	Pabst et al. 2025
Nectin-4	PD-L1 / angiogenic tumors	Novel targets being tested via imaging trials; potential for BgRT guidance in immuno-oncology	Kaur et al. 2025



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Limitations and Challenges

- Treatment session duration and patient tolerance
- Limited PET image resolution and system latency
- Need for robust automation and advanced QA tools
- Demonstration of clinical benefit and cost-effectiveness

	Radiotherapy— predicate studies	Idea	Development	Exploration	Assessment	Long-term evaluation
	Stage 0	Stage 1	Stage 2a	Stage 2b	Stage 3	Stage 4
Purpose	Answer the following questions before actual use of the innovation for treatment of patients	First time use of the innovation for treatment delivery in men	Technical optimization of the innovation for treatment delivery	Provide proof of early clinical effectiveness and safety of the innovation	Formal comparison of innovation against standard treatment	Long-term outcomes of the innovation, post-marketing, and surveillance

R-IDEAL framework for evaluating innovations in radiation oncology (this figure was cited from Verkooijen HM, et al. Front Oncol. 2017 Apr; 7: 59).

Summary of RefleXion PET-Linac online adaptive strategy

- First biology-driven adaptive RT
 - **Less about better “adaptive planning”**
 - More about: **real-time motion + biology-driven adaptation**
- **Not a replacement** for CBCT/MR adaptive workflows yet but a **parallel paradigm with strong future potential**

RefleXion PET-Linac: *Treat what lights up biologically, in real time*

MR-Linac: *See anatomy (soft tissue) continuously and adapt accordingly*

Ethos: *Replan based on daily CBCT efficiently*



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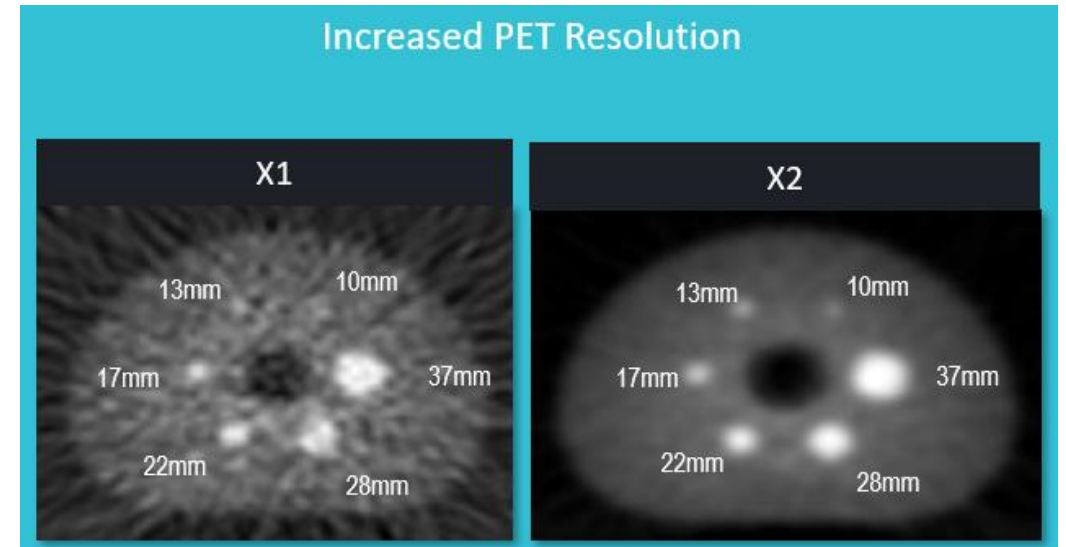
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Next two years

- **RefleXion X2 platform**

- Enabled by:
 - **20× PET sensitivity**
 - **20 cm FOV for motion coverage**
 - Aiming to increase 20% dose rate
→ improve total delivery time

- Aiming to handle **multi-lesion BgRT**
- Diagnostic-quality **16-slice kVCT + BgRT** → enabling next-generation hybrid adaptive radiotherapy



Future perspective

- Novel Radiotracer Development

- New radiotracers target hypoxia, proliferation, and immune markers, expanding PET-ART's tumor detection capabilities beyond FDG-avid lesions.
- Immune-targeted PET tracers visualize tumor-immune microenvironment, guiding immune-adaptive radiotherapy and combination treatment strategies.

- Expanding to multiple tumor sites

- Non-lung, Non-bone

- AI and Radiomics Integration

- Artificial intelligence and radiomics improve image interpretation, automate contouring, and enable real-time adaptive radiotherapy decision-making.

- Further enhanced PET Hardware/Software

- Advances in PET hardware enable faster, higher-resolution imaging, reducing latency and supporting true real-time treatment replanning.

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- Ann Jasman, RTT



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